Module:	CSC211 (2026)	Assessment:	Term 2 Practical 1A
Lecturer:	Mr. C. K. Baker	Total:	Pass / Fail
Duration:	Mon 14h00 – 16h30		

Topic
Binary Trees, Generics, OOP

Instructions

There are two (2) questions in this assignment. Submit a single compressed (.zip) file with all your code. Any sources used should be cited as comments in your code. The submission file should be named XXYYZZZ.zip where XXYYZZZ corresponds to your student number.

Question 1: Generic Binary Tree

Study the class definitions below, and answer the questions that follow.

```
class BinaryTreeNode<E>
{
    E value;
    BinaryTreeNode<E> left;
    BinaryTreeNode<E> right;
}

class BinaryTree<E>
{
    int size;
    BinaryTreeNode<E> root;
}
```

1.1 Copy the given code into two corresponding java files.

1.2 Use encapsulation to hide the attributes of `BinaryTreeNode<E>` and `BinaryTree<E>`. Each attribute should be visible only in its own class.

1.3 Write code for the following:

- a generic default constructor for each class
- a generic loaded constructor for each class
- generic getter methods for the attributes in each class
- generic setter methods for the attributes in each class

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Question 2: Instantiation and Testing

2.1 In a new file, `Main.java`, define and instantiate two (2) `BinaryTree` objects that correspond to the following binary trees when printed in level order (top-to-bottom, left-to-right):

- `a',`b',`c',`d',`e'
- 3.4, -1.5, 2.9, -9.3

2.2. Use the following `BinaryTree` object names:

- `bt1`
- `bt2`

Hint

Ensure each node in the tree is connected via appropriate node relationships. If no left child or right child exists, the value stored should be null.